

Solution to the Daily Limit

Based on the problem provided in the file `dailyintegral-limit-limit-1x1-2026-08-22.jpg`, we are tasked with finding the limit:

$$\lim_{x \rightarrow 0} \frac{1}{x^3} \left(\ln \left(\prod_{n=1}^{\infty} \frac{\Gamma\left(1 + \frac{x}{n}\right)}{e^{\frac{\pi^2 x^2}{12n^2} - \frac{\gamma x}{n}}} \right) \right)$$

where γ is the Euler-Mascheroni constant, $\Gamma(z)$ is the gamma function, and $\zeta(s)$ represents the Riemann zeta function.

1. Simplifying the Product and Logarithm

Using the properties of logarithms, we can convert the infinite product inside the expression into an infinite sum:

$$S(x) = \sum_{n=1}^{\infty} \left[\ln \Gamma\left(1 + \frac{x}{n}\right) - \left(\frac{\pi^2 x^2}{12n^2} - \frac{\gamma x}{n} \right) \right]$$

2. Taylor Series Expansion of the Gamma Function

Recall the Taylor series expansion of $\ln \Gamma(1 + z)$ around $z = 0$:

$$\ln \Gamma(1 + z) = -\gamma z + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} z^k$$

Writing out the first few terms explicitly and using $\zeta(2) = \frac{\pi^2}{6}$, we have:

$$\ln \Gamma(1 + z) = -\gamma z + \frac{\pi^2}{12} z^2 - \frac{\zeta(3)}{3} z^3 + \mathcal{O}(z^4)$$

Substituting $z = \frac{x}{n}$ into the expansion yields:

$$\ln \Gamma\left(1 + \frac{x}{n}\right) = -\gamma \frac{x}{n} + \frac{\pi^2 x^2}{12n^2} - \frac{\zeta(3)x^3}{3n^3} + \mathcal{O}\left(\frac{x^4}{n^4}\right)$$

3. Evaluating the Series Summation

Next, substitute this expansion back into the terms of $S(x)$:

$$\begin{aligned} \ln \Gamma\left(1 + \frac{x}{n}\right) - \left(\frac{\pi^2 x^2}{12n^2} - \frac{\gamma x}{n} \right) &= \left(-\gamma \frac{x}{n} + \frac{\pi^2 x^2}{12n^2} - \frac{\zeta(3)x^3}{3n^3} + \mathcal{O}\left(\frac{x^4}{n^4}\right) \right) - \left(\frac{\pi^2 x^2}{12n^2} - \frac{\gamma x}{n} \right) \\ &= -\frac{\zeta(3)x^3}{3n^3} + \mathcal{O}\left(\frac{x^4}{n^4}\right) \end{aligned}$$

Summing this expression from $n = 1$ to ∞ :

$$\begin{aligned} S(x) &= \sum_{n=1}^{\infty} \left(-\frac{\zeta(3)x^3}{3n^3} + \mathcal{O}\left(\frac{x^4}{n^4}\right) \right) \\ &= -\frac{\zeta(3)x^3}{3} \sum_{n=1}^{\infty} \frac{1}{n^3} + \mathcal{O}(x^4) \\ &= -\frac{\zeta(3)^2}{3} x^3 + \mathcal{O}(x^4) \end{aligned}$$

since $\sum_{n=1}^{\infty} \frac{1}{n^3} = \zeta(3)$.

4. Evaluating the Limit

Now, substitute $S(x)$ back into the original limit expression:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{1}{x^3} \left(\ln \left(\prod_{n=1}^{\infty} \frac{\Gamma\left(1 + \frac{x}{n}\right)}{e^{\frac{\pi^2 x^2}{12n^2} - \frac{\gamma x}{n}}} \right) \right) &= \lim_{x \rightarrow 0} \frac{1}{x^3} \left(-\frac{\zeta(3)^2}{3} x^3 + \mathcal{O}(x^4) \right) \\ &= -\frac{\zeta(3)^2}{3}\end{aligned}$$

Conclusion

Therefore, the evaluated limit is:

$$-\frac{\zeta(3)^2}{3}$$